

AI surrogate models for forecasting tides and storm-surges

Martin Verlaan^{1 2},

Jing Zhao^{1 2},

Willem Tromp¹,

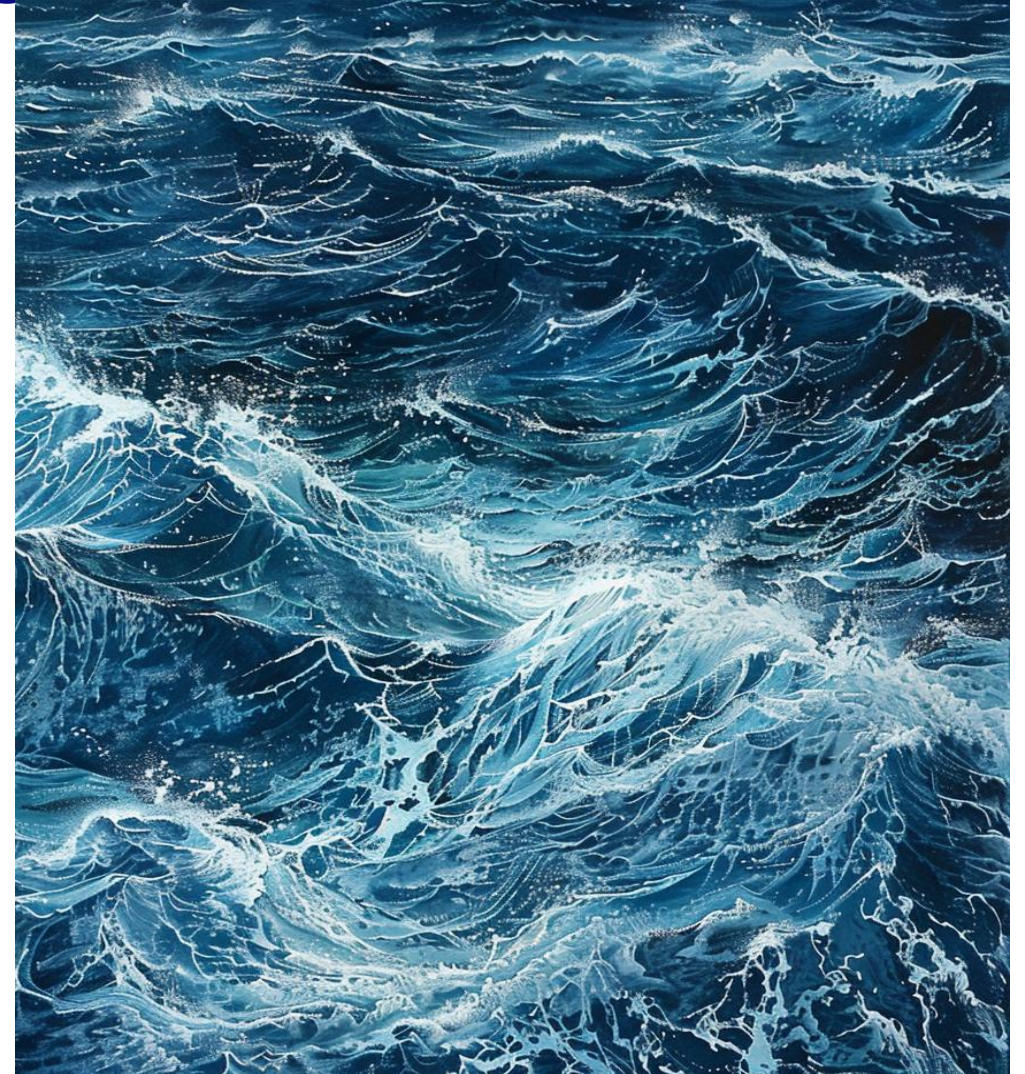
Yilin Tang²,

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¹Department of Hydrodynamics and Forecasting, Deltares

²Delft Institute of Applied Mathematics, TU Delft

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Outline

- Why a surrogate for tides and storm-surges?
- A time-series approach: enforce some physics
- A GNN based model: accumulation of errors for unrolling
- Example guideline for stability
- Example design issues
- Challenges

Motivation – Traditional numerical models & challenges

- Deltares has developed:
 - Global Tide and Surge Model (GTSM)
 - Dutch Continental Shelf Model – Flexible Mesh (DCSM-FM)
- Delft3D Flexible Mesh software
- **Challenge:** High computational cost

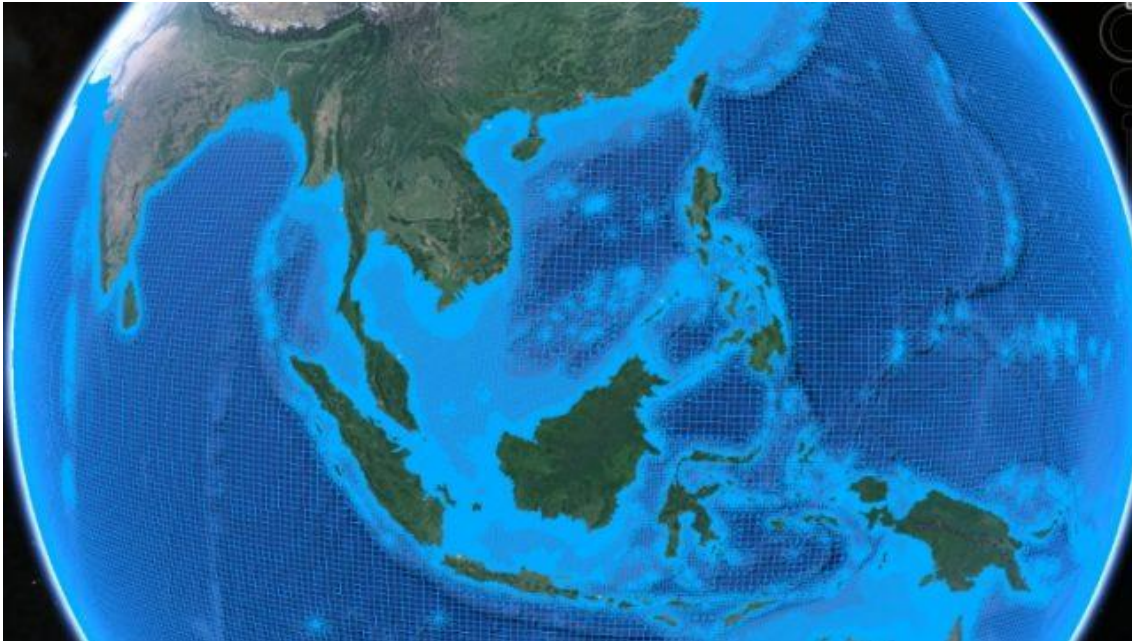


Figure: Grid of the Global Tide and Surge Model for Southeast Asia, from <https://www.deltares.nl/en/expertise/projects/global-modelling-of-tides-and-storm-surges>

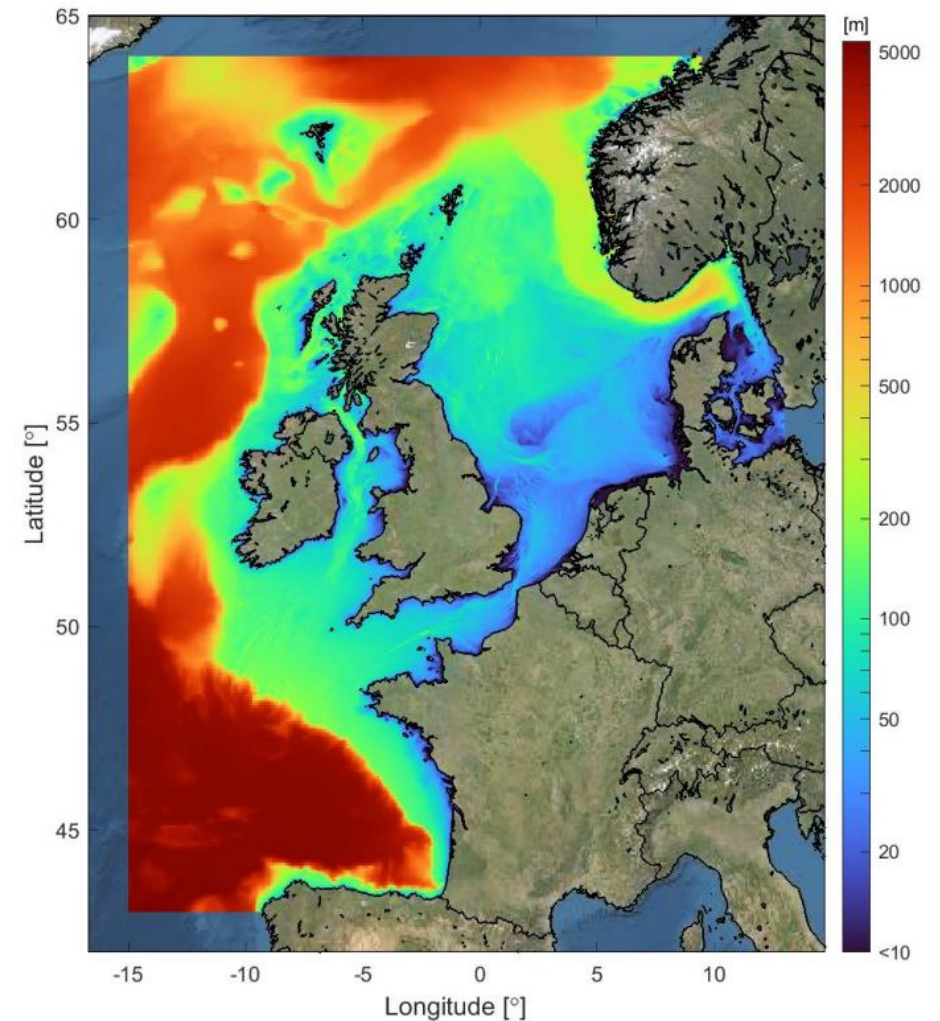
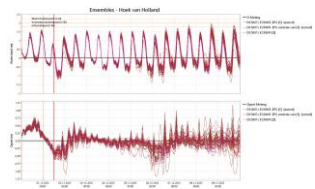


Figure: Overview of the DCSM-FM model bathymetry, from Zijl, F., et al. *3D DCSM FM: a sixth-generation model for the NW European Shelf*. Tech. rep., Deltares, available online at: https://publications.deltares.nl/11208054_004_0003.pdf

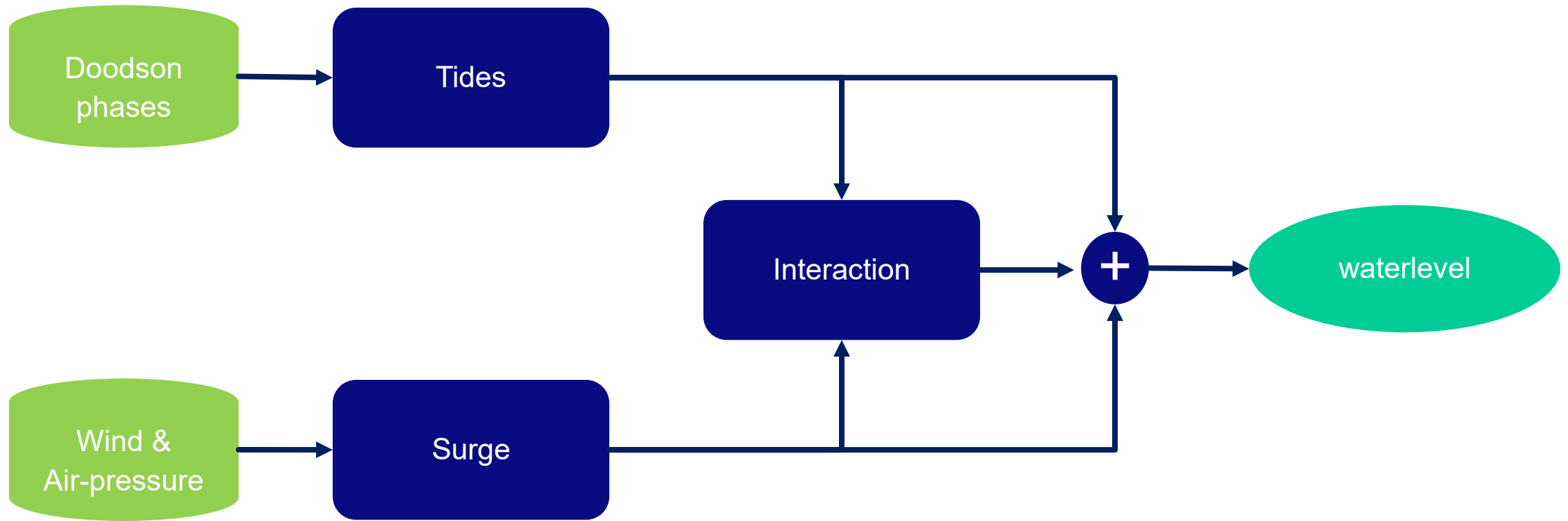
Ensemble Prediction



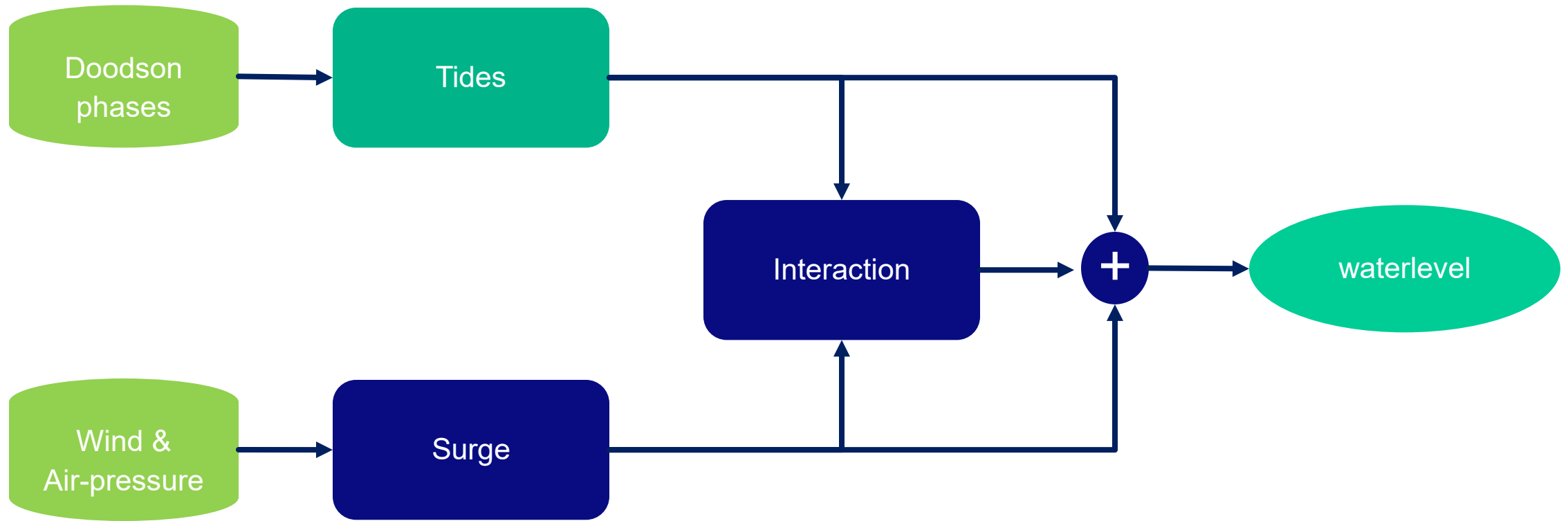
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| Time-series ML model

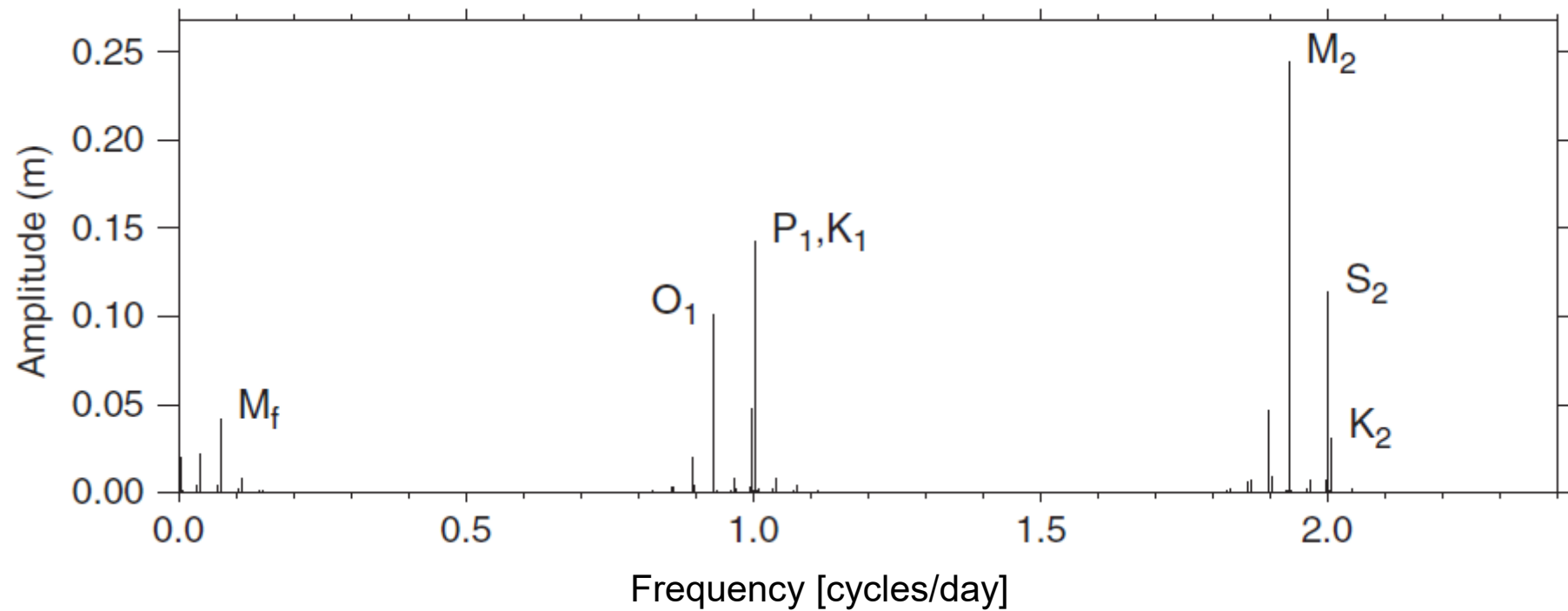
Waterlevel=tides+surge+interaction



Waterlevel=tides+surge+interaction



Tidal frequencies



Doodson phases and tidal analysis

Table 3.2 Basic astronomical periods and frequencies

	Period		Frequency		Angular speed	
			<i>f</i> cycles per mean solar year	<i>σ</i> degrees per mean solar hour	Symbol in rate of radians	Rate of change of
Mean solar day	1.00	mean solar days	1.00	15.0000	ω_0	C_s
Mean lunar day	1.0351	mean solar days	0.9661369	14.4921	ω_1	C_l
Sidereal month	27.3217	mean solar days	0.0366009	0.5490	ω_2	s
Tropical year	365.2422	mean solar days	0.0027379	0.0411	ω_3	h
Moon's perigee	8.85	Julian years	0.0003093	0.0046	ω_4	p
Regression of Moon's nodes	18.61	Julian years	0.0001471	0.0022	ω_5	N
Perihelion	20,942	Julian years	–		ω_6	p'

$$\omega = \sum_{i=1}^6 d_i \omega_i$$

$$\psi_i = (d_i \omega_i)(t - t_0) + \phi_i$$

Compound tides and over-tides

$$\partial_t h + \partial_x(Hu) + \partial_y(Hv) = 0$$

$$\partial_t(Hu) + \partial_x(Hu^2) + \partial_y(Huv) - fHv + gH\partial_x h + ku|(u, v)| = \tau_x/\rho - H\partial_x p_a$$

$$\partial_t(Hv) + \partial_x(Huv) + \partial_y(Hv^2) + fHu + gH\partial_y h + kv|(u, v)| = \tau_y/\rho - H\partial_y p_a$$

Non-linear interactions

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}(\cos(\alpha + \beta) + \cos(\alpha - \beta))$$

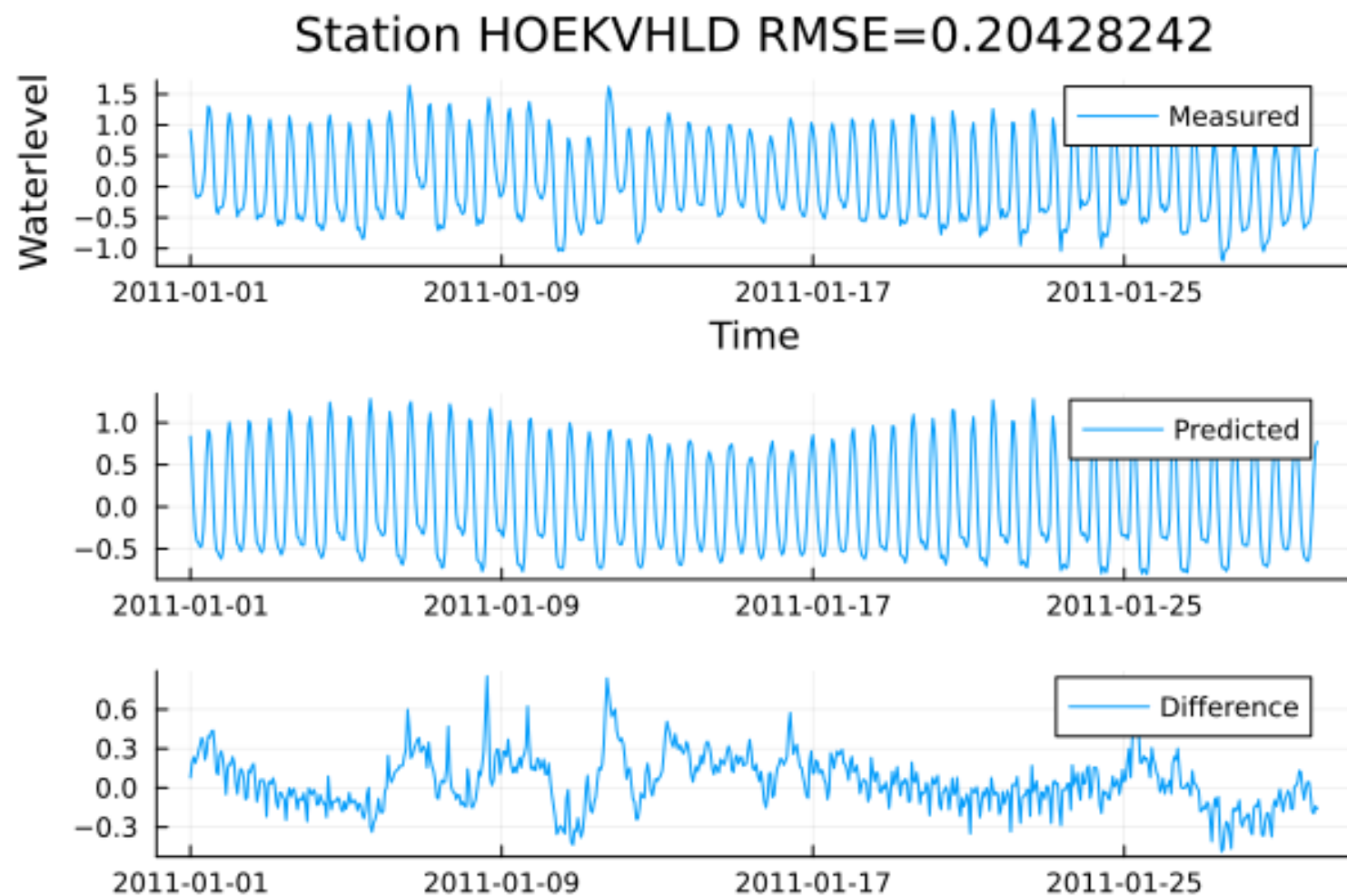
$$\cos(\alpha)\sin(\beta) = \frac{1}{2}(\sin(\alpha + \beta) - \sin(\alpha - \beta))$$

→ Include multiplication in network layer

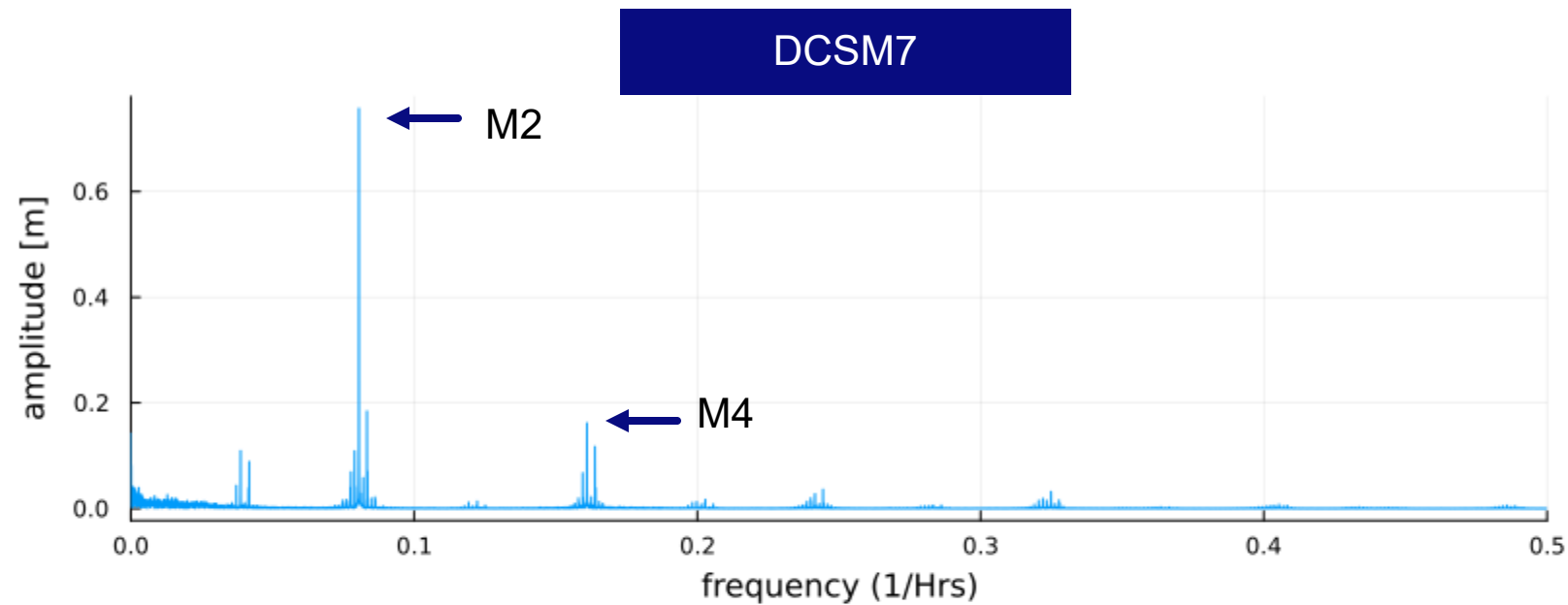
Example

Name	Doodson	Degrees / hour
M2	2 0 0 0 0 0	28.9841
S2	2 2 -2 0 0 0	30.0000
MS4	4 2 -2 0 0 0	58.9841

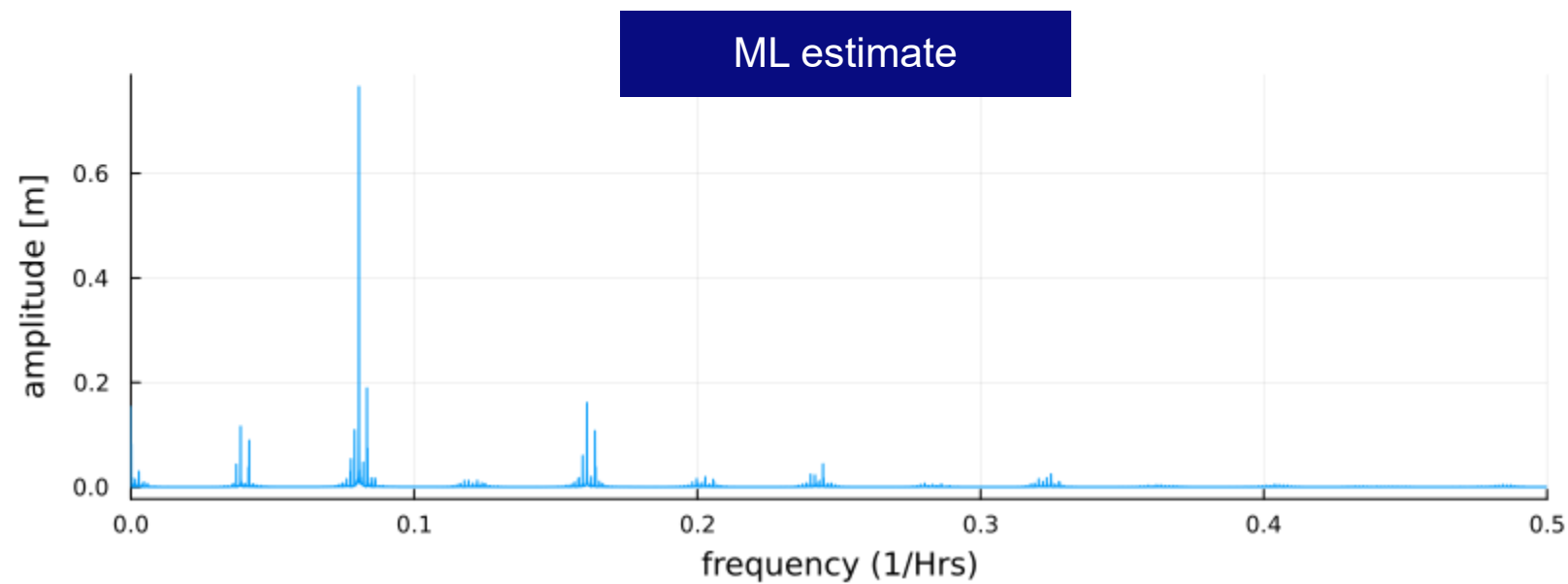
Tide results test



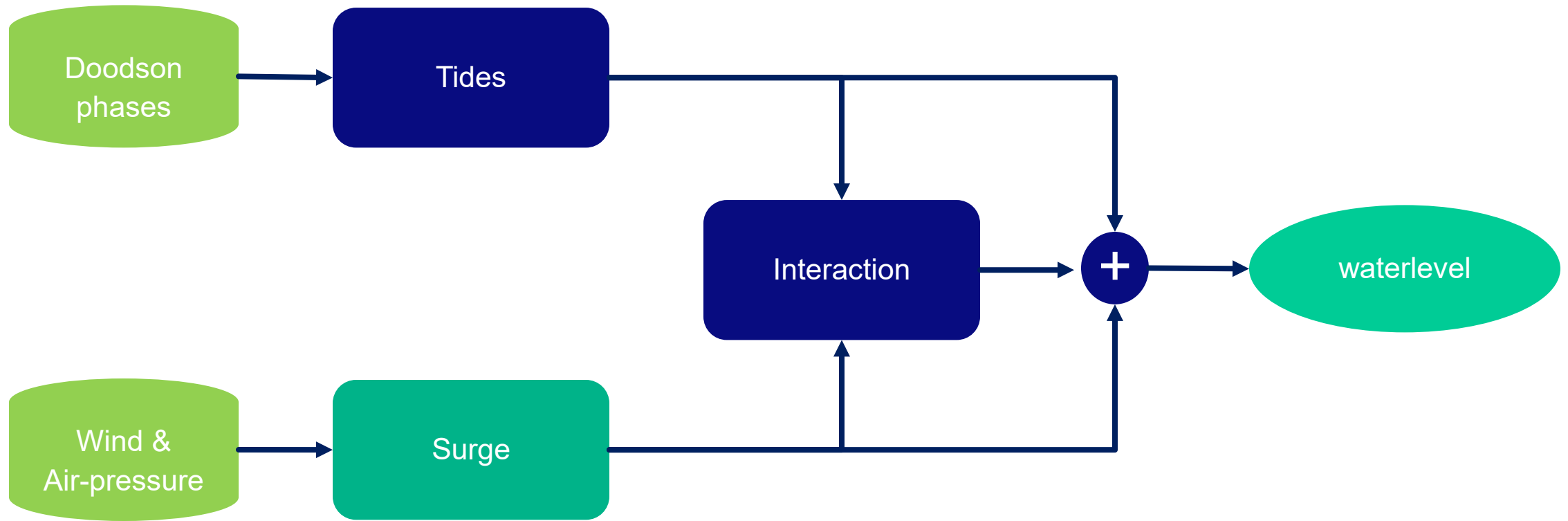
Frequencies



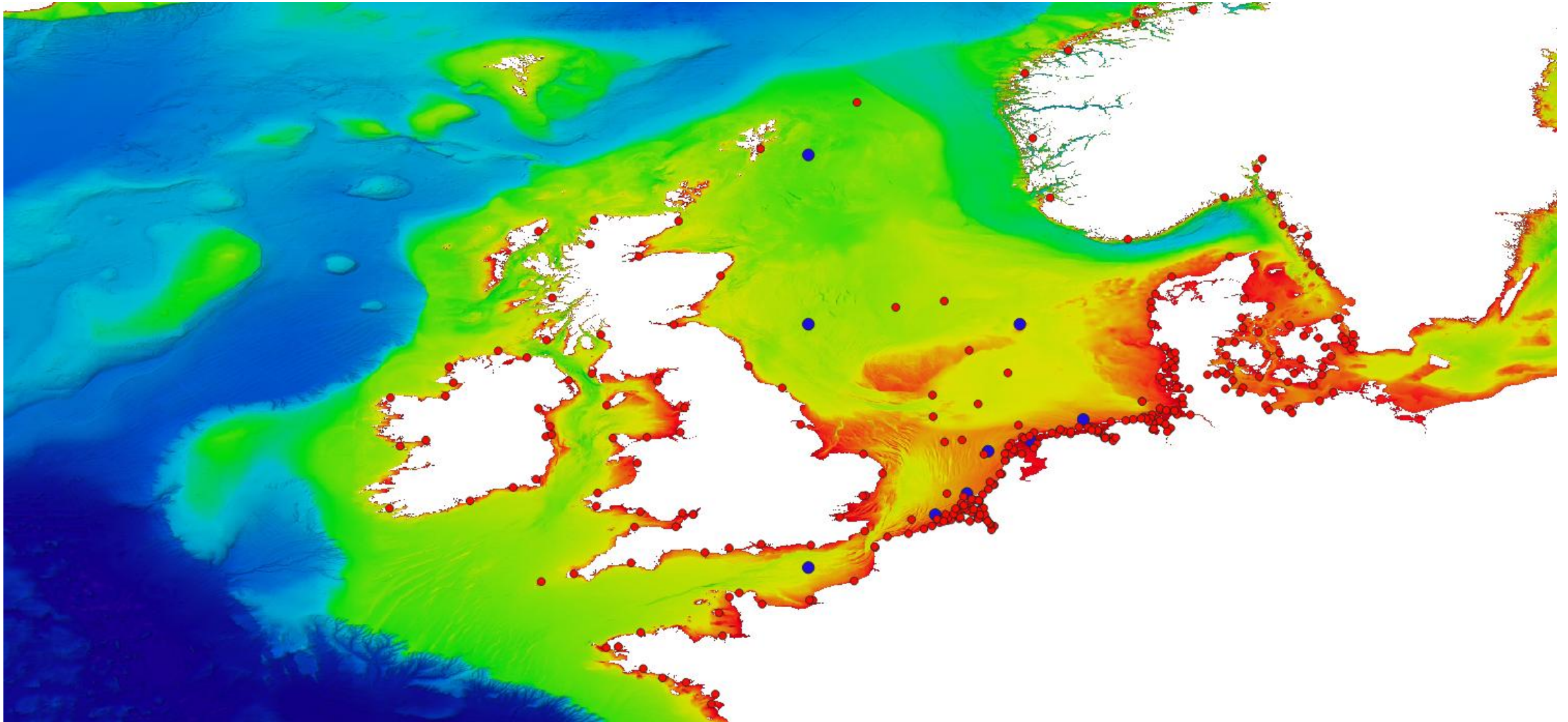
Many tidal frequencies are generated by the network



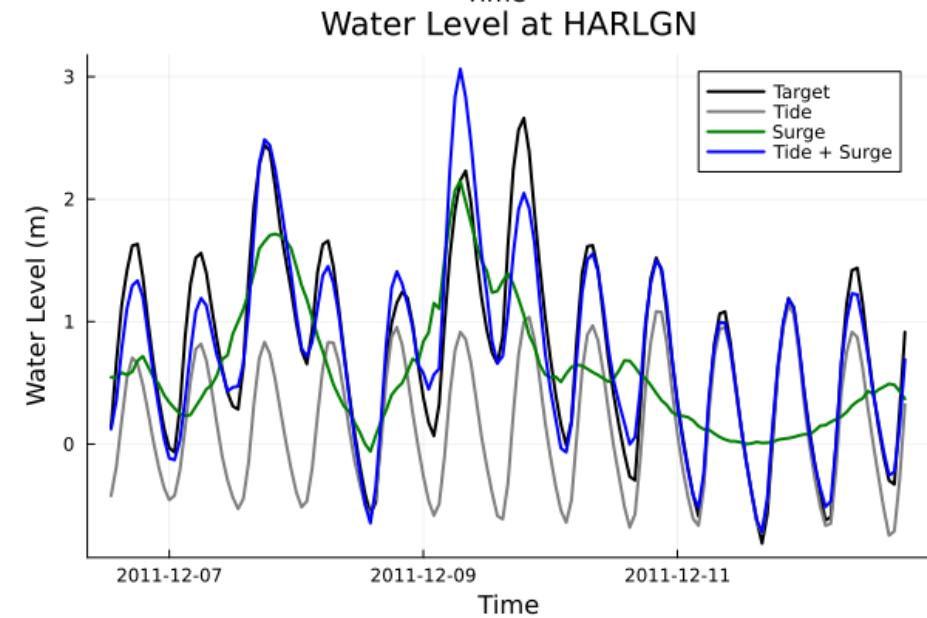
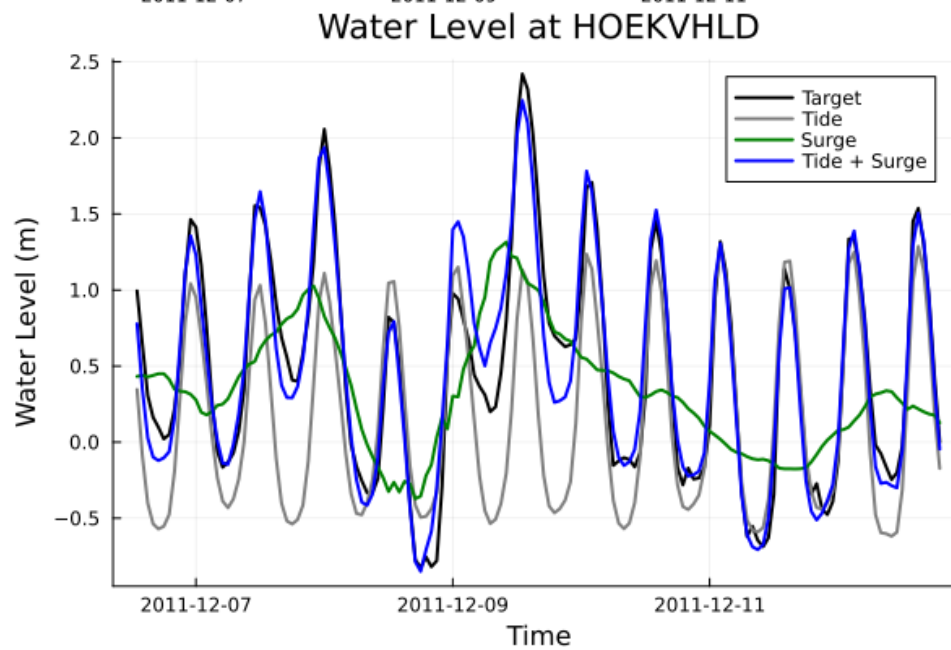
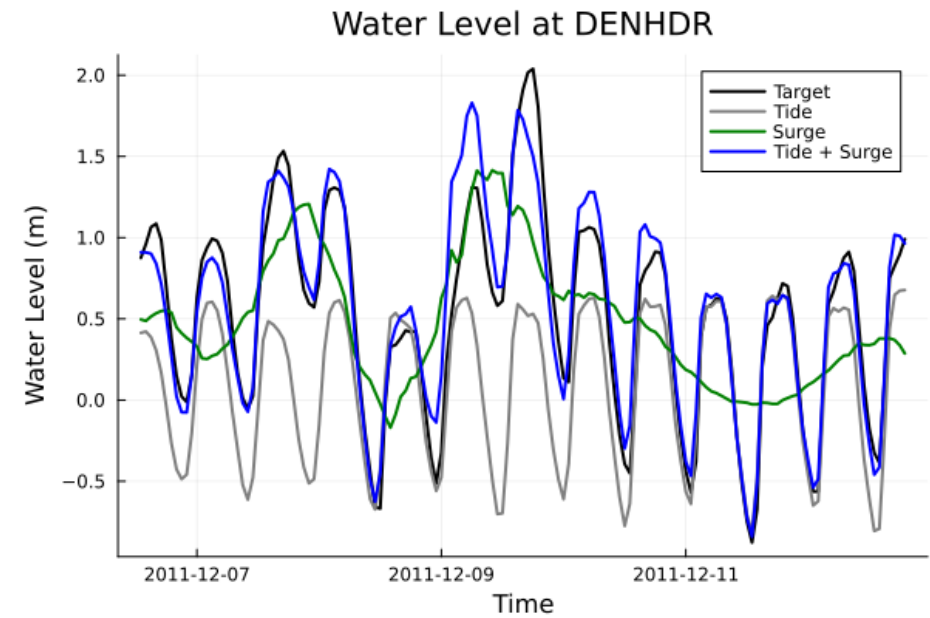
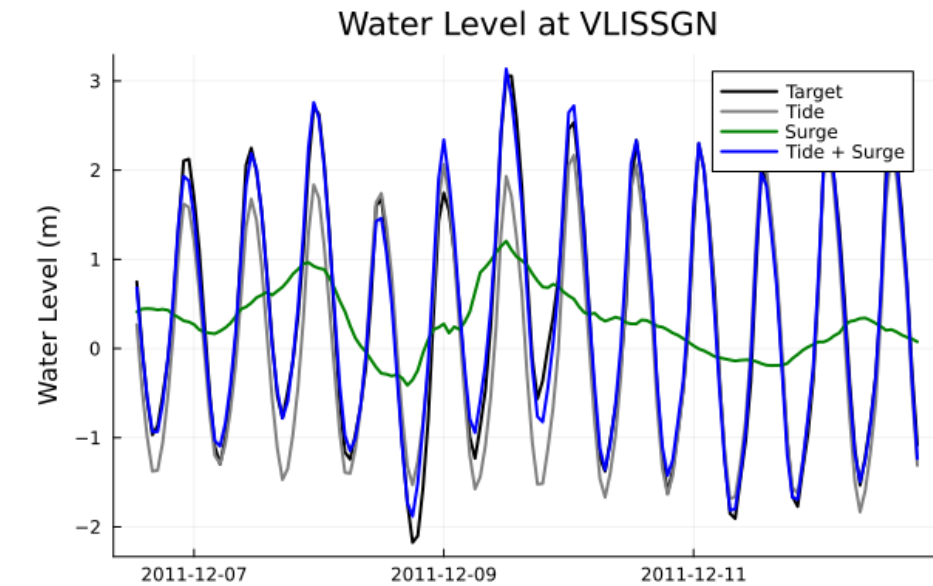
Waterlevel=tides+surge+interaction



Storm-surge locations

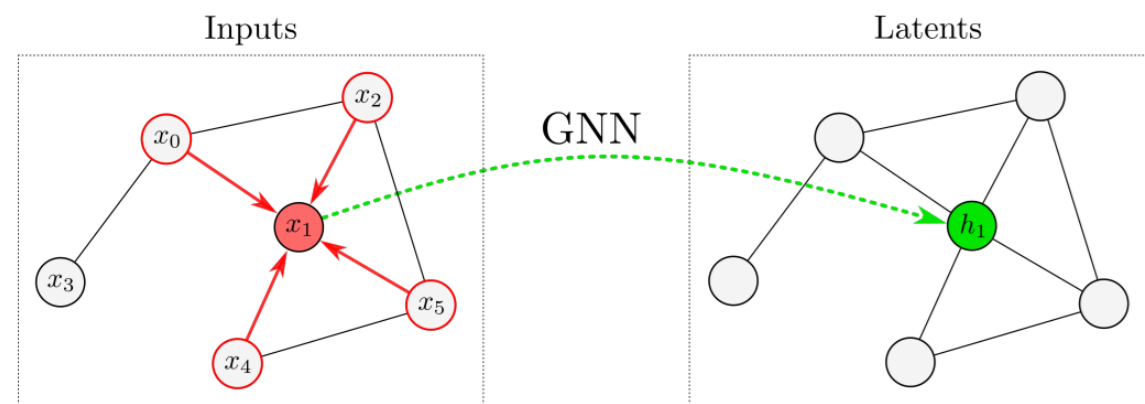
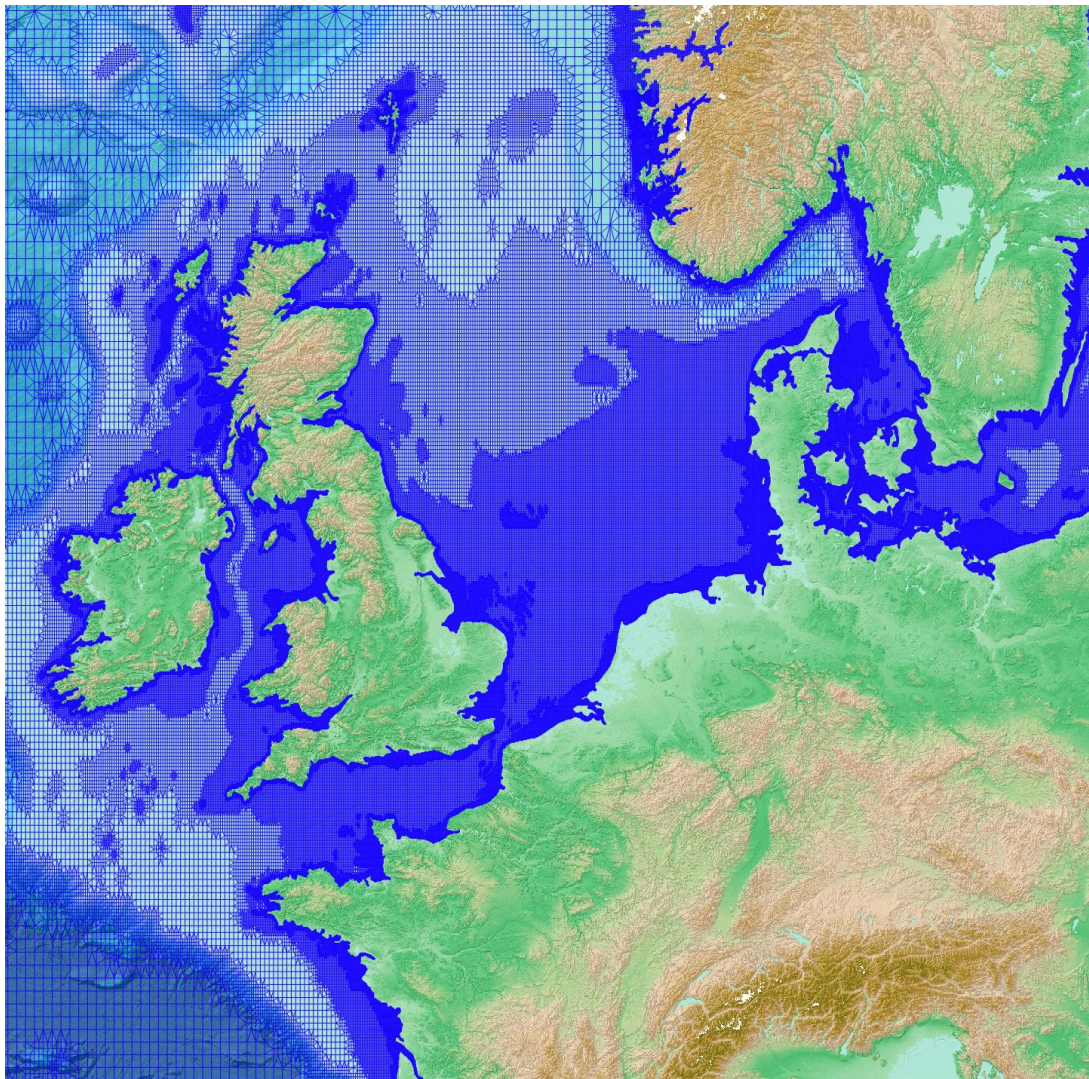


Test storm

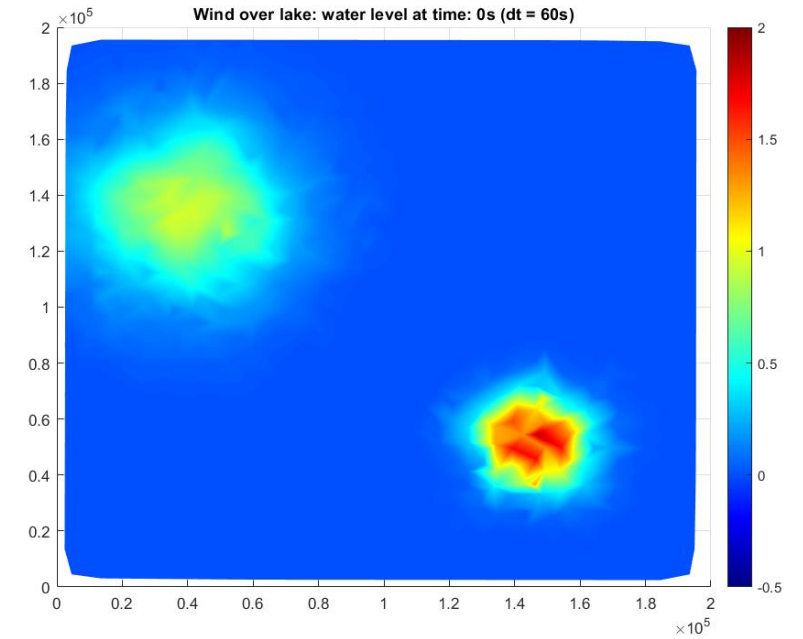
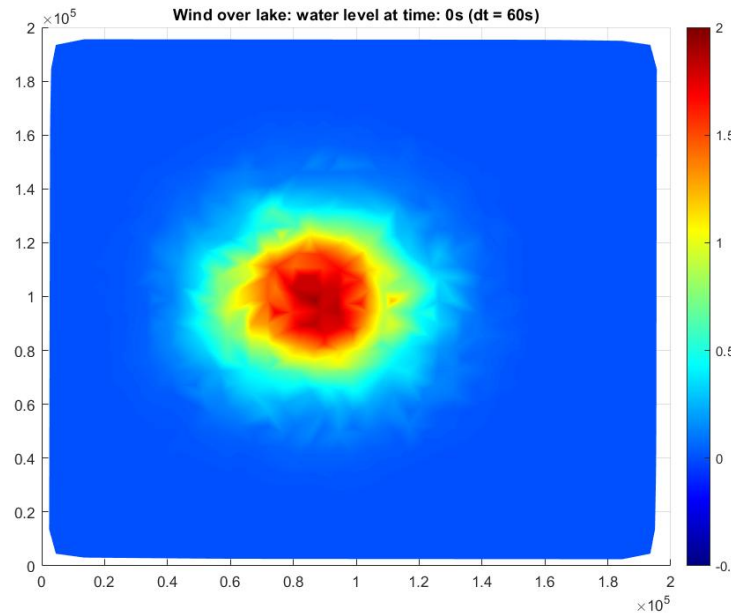
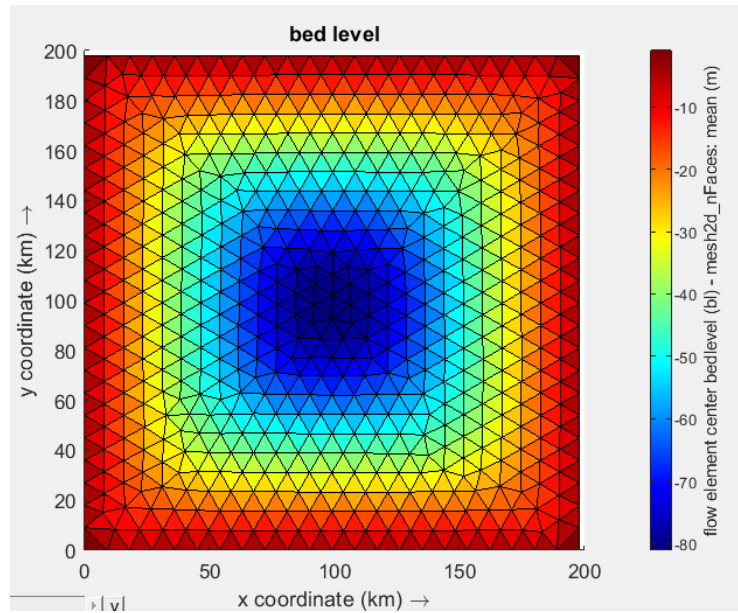


| GNN surrogate ML model

A Graph Neural Network for tides & storm-surges



Test case: Lake



Training data:

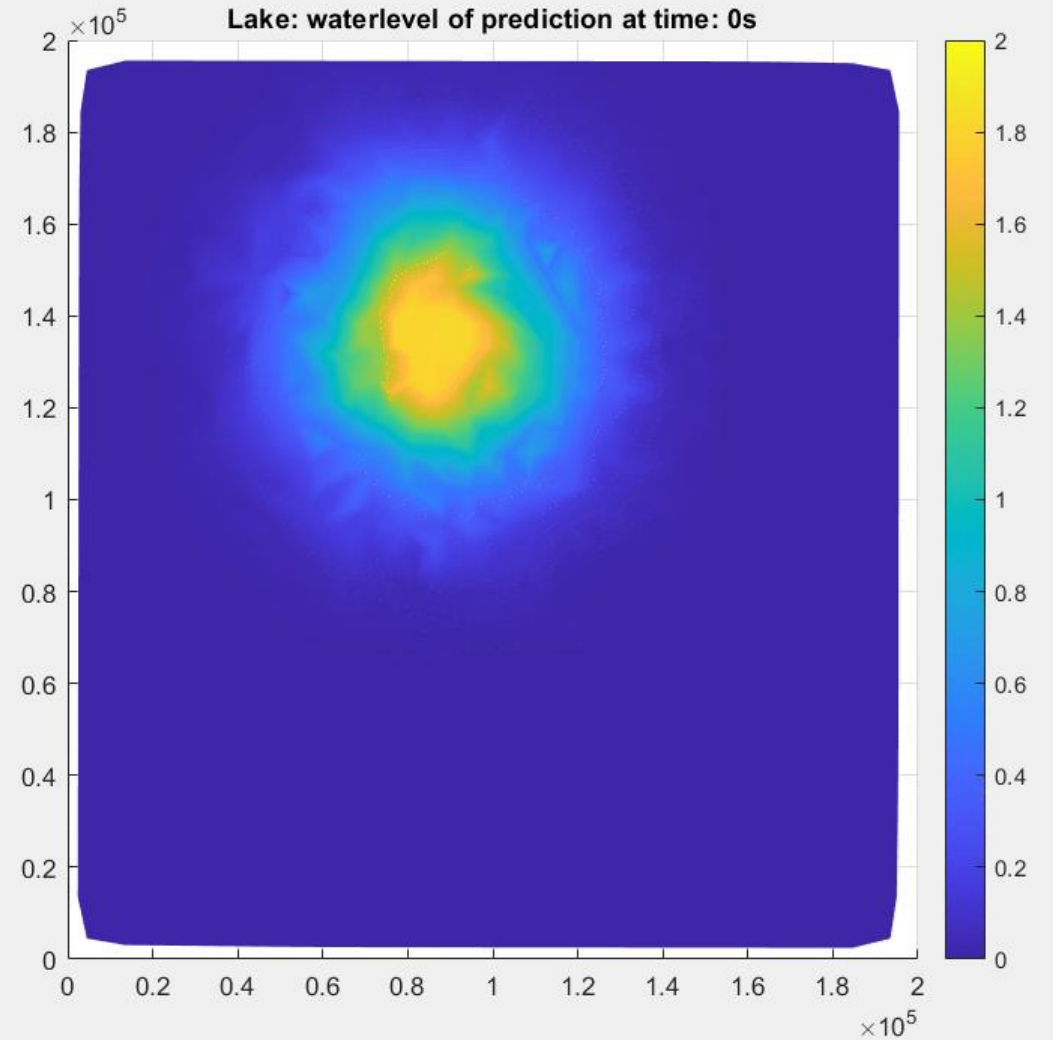
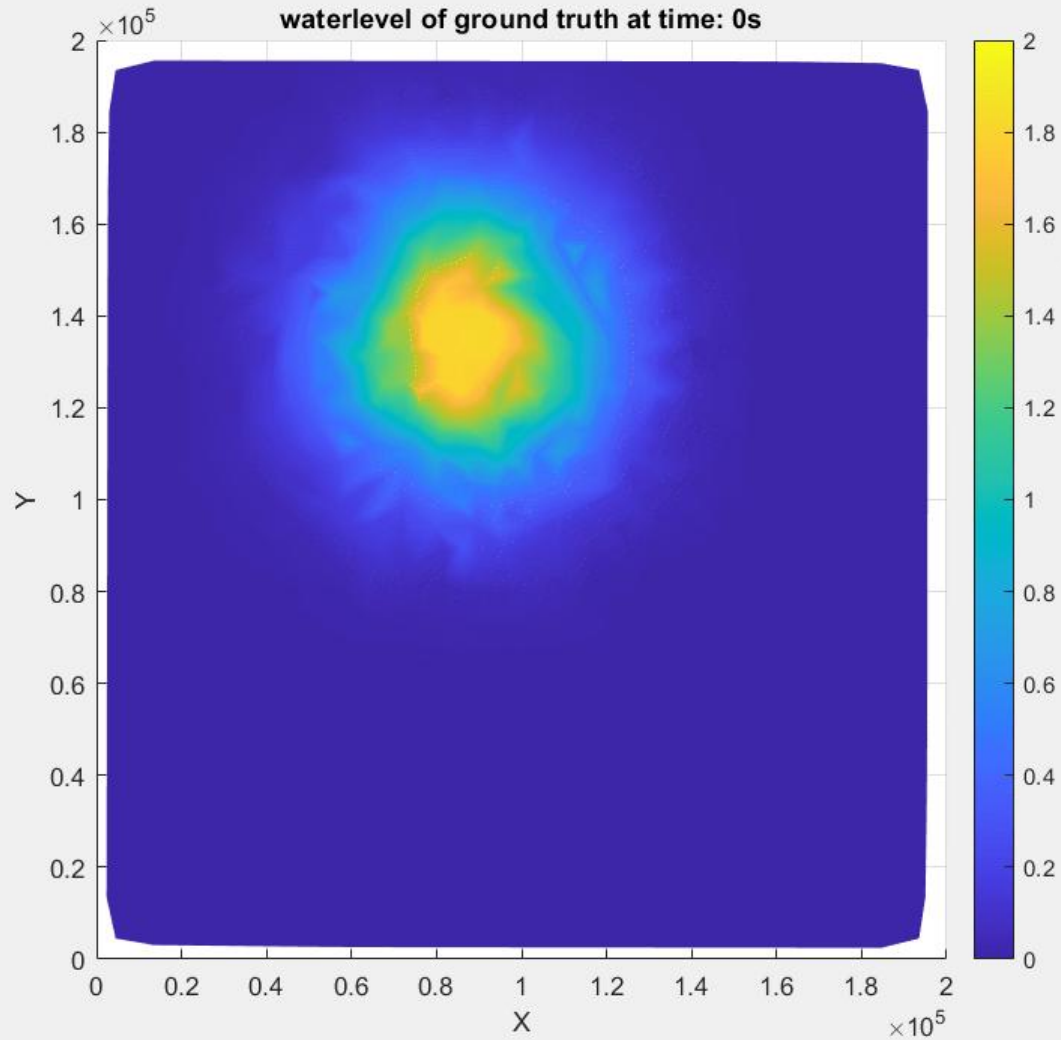
- 200 simulations with different initial water levels:
 - 1 – 100: 1 bump,
 - 101-200: 2 bumps:
- Each simulation:
200min, dt = 1min

GNN model

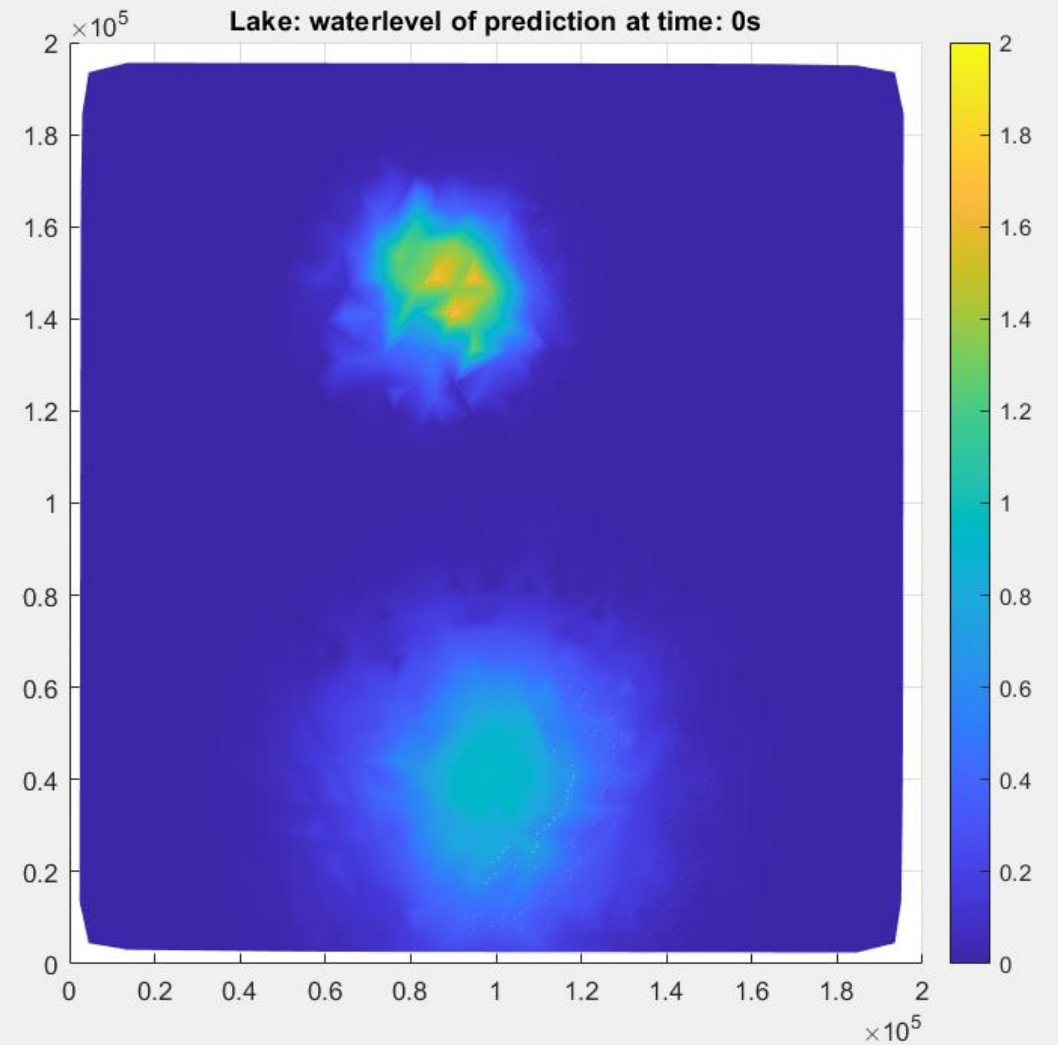
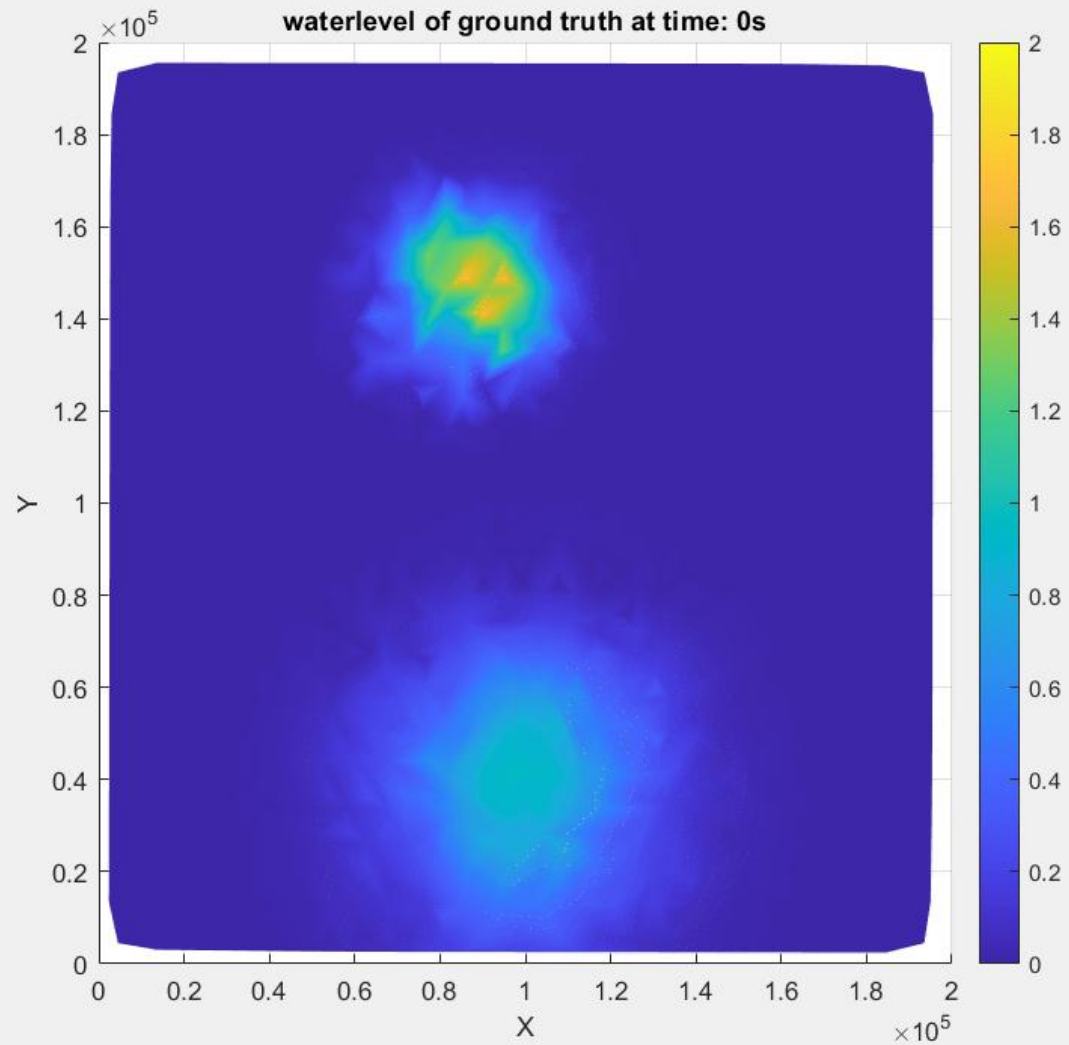
- Message passing = 7
- Layer_size = 128
- hidden_layer = 2
- Batch = 1
- Noise: Gauss distributed, with standard deviation 0,02
- Adam optimizer
- Learning_rate: from 0.005 to 0,0001

Data files	train	valid	test
Number of simulations	140	40	20

Test case: results (1)



Test case: results (2)



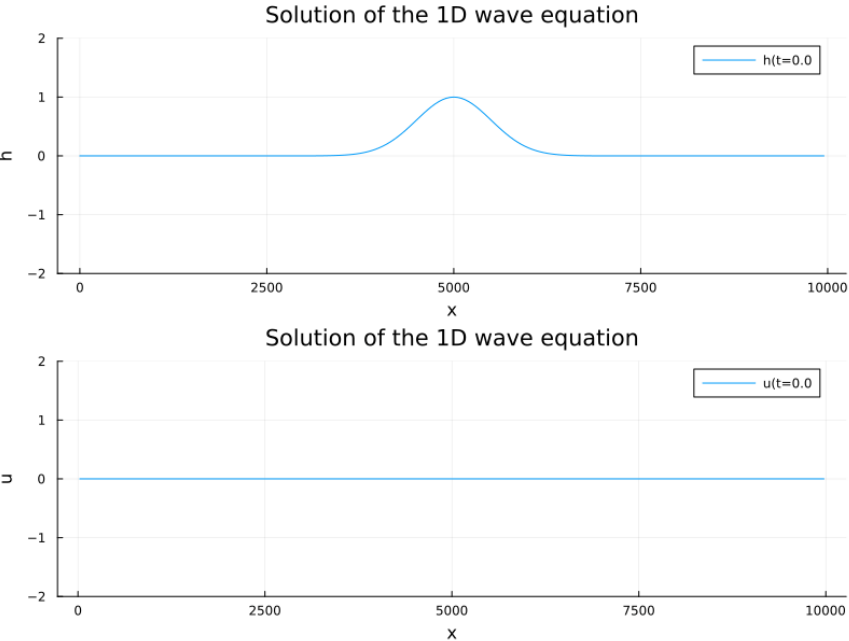
CFL condition

$$\frac{\partial h}{\partial t} + D \frac{\partial u}{\partial x} = 0$$
$$\frac{\partial u}{\partial t} + g \frac{\partial h}{\partial x} = 0$$

$$x \in [0, 10000], D = g = 10.$$

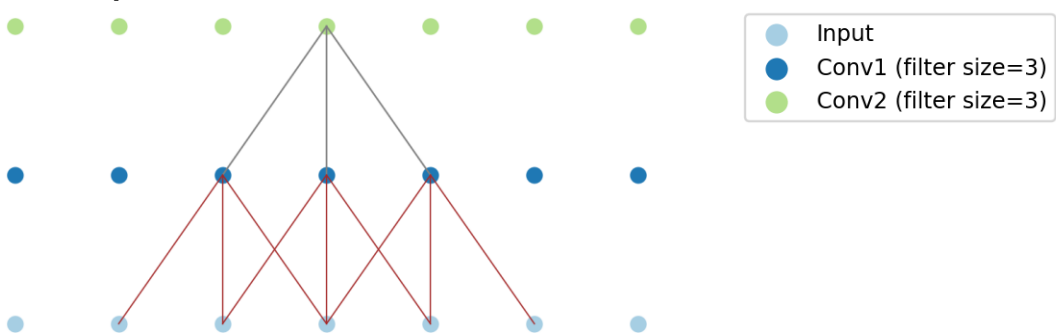
$$t \in [0, 2000], \Delta x = 50, \Delta t = 5.0$$

$$h(x, t = 0) = \exp\left(-\frac{(x - 0.5)^2}{2w^2}\right), \quad w = 0.03 : 0.01 : 0.2$$
$$u(x, t = 0) = 0$$



Wave equation

Receptive field



Stability experiments

n_layers\CNN time step	5	10	15	20	25	30
3			×	×	×	×
4				×	×	×
5				×	×	×
6					×	×

n_layers\CNN time step	5	10	15	20	25	30	35	40	45	50
3						×	×	×	×	×
4							×	×	×	×
5	×								×	×
6	×	×								×

Conclusion: CNN is also bound by CFL condition

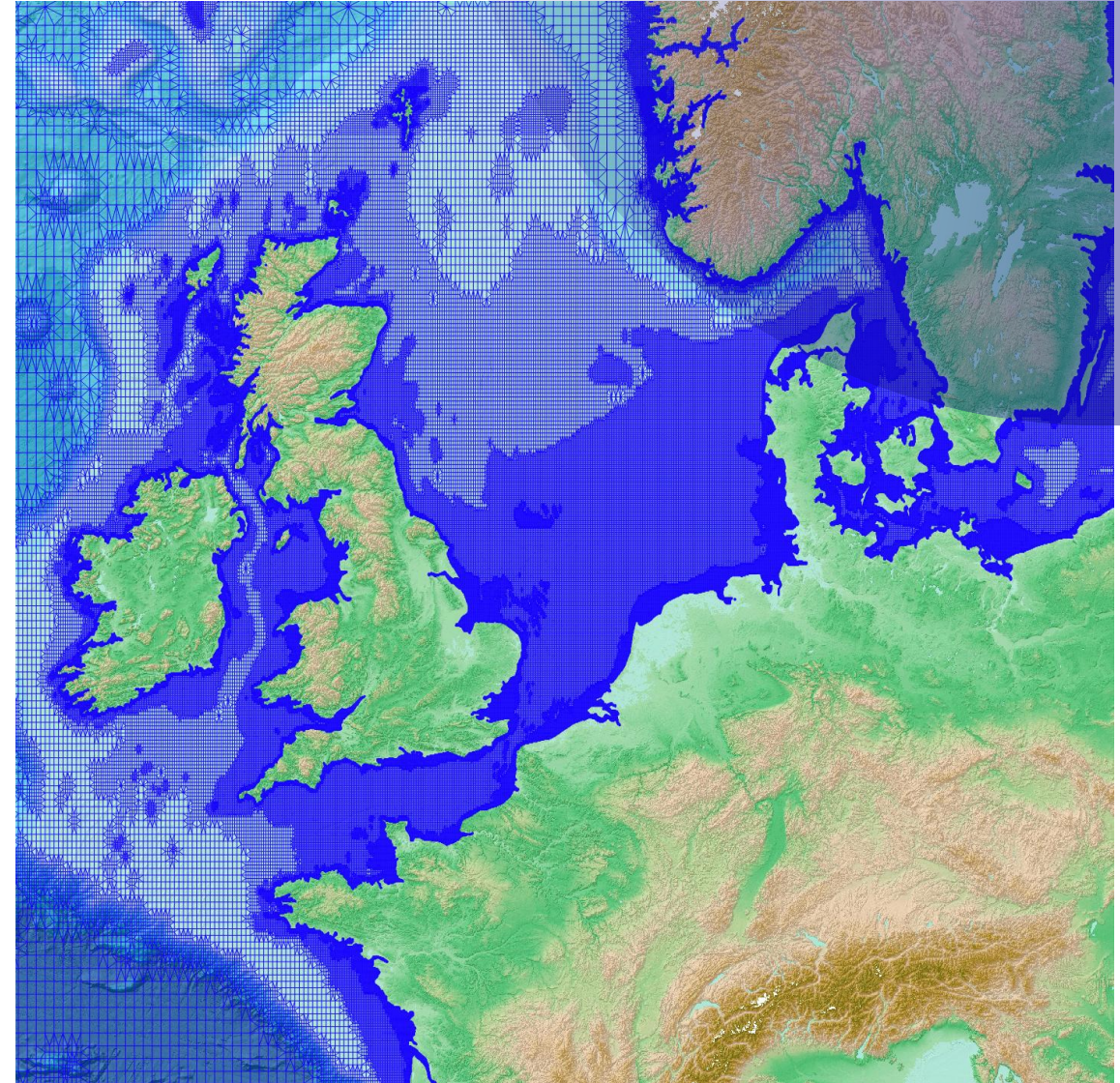
Challenges (for discussion)

A surrogate model should approximate the source model with sufficient accuracy

Development is largely experimental and based on heuristics

There are very few methods for analysis

- To pinpoint the weak points
- To show the limits of what's possible



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Thanks for your attention.
Questions?



Deltares



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